Machine Learning

Multi Layer Perceptron & Unsupervised Learning

DPP : 01

Q1 In a semi-supervised learning dataset, if there are 1000 examples and only 100 examples are labeled, what is the ratio of labeled to unlabeled examples?

Q2 Which of the following statements regarding bias and variance are true?

(A) Models which overfit have a high bias and underfit have a high variance.

(B) Models which overfit have a high bias and underfit have a low variance.

(C) Models which overfit have a low bias and underfit have a high variance.

(D) Models which overfit have a low bias and underfit have a low variance.

Q3 Given the distance matrix below, which pair of points will be merged first in single linkage clustering?

	x_1	x_2	x_3	x_4
x_1	0	4	3	6
x_2	4	0	5	7
x_3	3	5	0	8
x_4	6	7	8	0

(A) (x_1, x_2) (B) (x_2, x_3)

(C) (x_1, x_3) (D) None of these

Q4 Which of the following represents the linkage criteria of hierarchical clustering?

(A) The distance metric used for measuring dissimilarity between clusters.

(B) The method used to initialize cluster centroids.

(C) The number of iterations required for convergence.

(D) The threshold used for merging clusters.

Q5 How does a feedforward neural network differ from other types of neural networks?

(A) Feedforward neural networks allow feedback connections between neurons.

(B) Feedforward neural networks can only have one layer of neurons.

(C) Feedforward neural networks do not contain any hidden layers.

(D) In feedforward neural networks, information flows in one direction, from input to output, without cycles or loops.

Q6 Given an input layer with three nodes x_1 x_2, and x_3, having value 0.7,0.5 and 0.3 respectively, associated with weights 0.2,0.3 and -0.2. If the bias is 0.48 and a binary sigmoid activation function is used, find the output of the neural network? (Rounded upto 2 decimal)

Q7 You want to cluster 8 observations into 4 clusters using the K-Means clustering algorithm. After the first iteration, the clusters C_1,C_2,C_3 and C_4 have the following observations:

C_1 : {(2,3), (6,8) , (8,7))

C_2 : {(1,0) , (4,2)}

C_3: {(9,9) , (7,6)}

C_4: {(3,4) , (0,1)}

What will be the cluster centroids if you proceed to the second iteration?

(A) C_1: (5.3,6), C_2: (2.5,1), C_3: (8,7.5), C_4: (1.5,2.5)

(B) C_1 : (4.3,4), C_2 : (2,1.2), C_3 : (7.5,7.5), C_4 : (1.5,2.5)

(C) C_1:(7,8), C_2 : (3,1),C_3:(9,8),C_4 :(1,2)

(D) None of these

Q8 Which of the following describes what the property of unbiasedness means?

(A) The shape of the sampling distribution is approximately normally distributed.

(B) The center of the sampling distribution is found at the population standard deviation.

(C) The center of the sampling distribution is found at the population parameter that is being estimated.

(D) The sampling distribution in question has the smallest variation of all possible sampling

distributions.

Q9 Which of the following statements are true?

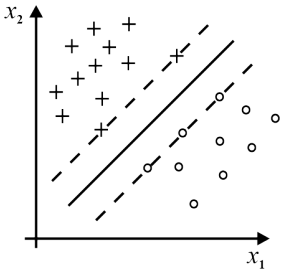
(A) The chances of overfilling decreases with increasing the number of hidden nodes and increasing the number of hidden layers.

(B) A neural network with one hidden layer can represent any Boolean function given sufficient number of hidden units and appropriate activation functions.

(C) Two hidden layer neural networks can represent any continuous functions (within a tolerance) as long as the number of hidden units is sufficient and appropriate activation functions used

(D) All of the above

Q10 What is the leave-one-out cross-validation error estimate for maximum margin separation in the following figure? (We are asking for a number)



(A) –1 (B) –2

(C) 0 (D) 1

Q11 Consider a CNN with an input feature map of size 8 × 8 and a Max Pooling layer with a pool size of 2 × 2 and a stride of 2. How many elements will be there in the output feature map after applying the Max Pooling operation?

(A) 16 (B) 12

(C) 9 (D) 4

Q12 Consider a simple neural network with one hidden layer having three neurons and an output layer with one neuron. The activation function used is the sigmoid function. The network is trained using back propagation. The input features are x_1 = 0.8, x_2 = 0.6 and the target output is ytarget = 0.9.

The weights and biases are initialized as follows:

- Hidden layer weights: = ω_{11} = 0.4, ω_{21} = 0.2, ω_{31} = –0.5
- Output layer weights: ω_{12} = 0.7, ω_{22} = 0.3, ω_{32} = –0.6
- Hidden layer biases: b_1 = 0.3, b_2 = –0.1 , b_3 = 0.2
- Output layer bias: b_2 = –0.2

Now, calculate the error term (delta) for the output neuron using the back propagation formula:

$\delta_{\text{output}} = (y_{\text{output}} - y_{\text{target}}) \cdot \sigma(z_{\text{output}})$

(A) δ_{doutput} = 0.122

(B) δ_{doutput} = –0.122

(C) δ_{doutput} = 0.018

(D) None of these

Q13 Which of the following statements is False regarding Leave-One-Out (LOO) Cross-Validation?

(A) In LOO-CV, for each iteration, a random subset of samples is selected as the validation set.

(B) LOO-CV involves training the model on all samples except one, and this process is repeated for each sample in the dataset.

(C) LOO-CV is computationally efficient and typically requires less processing power compared to other cross-validation methods.

(D) The primary advantage of LOO-CV is that it prevents overfitting by leaving out the same sample in each iteration.

Q14 Consider the Softmax function commonly used in the output layer of a neural network for multiclass classification. The Softmax function takes an input vector z and produces an output vector s(z) where each element s(zi) represents the probability of the corresponding class.

Which of the following statements does not describes a property of the Softmax function?

(A) The Softmax function always outputs probabilities in the range (0,1) [0,1].

(B) The Softmax function introduces non linearity to the neural network.
(C) The output probabilities from the Softmax function are independent of each other.
(D) The Softmax function is mainly used for regularization purposes.

Q15 Given below are two statements:

If two variables V_1 and V_2 are used for clustering, then consider the following statements for k means clustering with k = 3.

- Statement I: If V_1 and V_2 have correlation of 1 the cluster centroid will be in straight line.
- Statement II: If V_1 and V_2 are correlation of 0 the cluster centroid will be in straight line.

In the light of the above statements, choose the correct answer from the options given below:
(A) Statement I is correct
(B) Statement II is correct
(C) Statement I is wrong
(D) Statement II is wrong

Q16 Consider a neural network with a single neuron in the input layer and a single neuron in the output layer. The weights connecting these neurons are given as follows: $\omega_1 = -2$, $\omega_2 = -5$ and $\omega_3 = -4$
Given an input vector X= [2,3,1], perform the forward propagation to calculate the output of the neural network.
What will be the weighted sum of input?
(A) –23 (B) –46
(C) –20 (D) +21

Q17 After training a neural network, you observe a large gap between the training accuracy (100%) and the test accuracy (42%). Which of the following methods is commonly used to reduce this gap?
(A) Generative adversarial networks
(B) Dropout
(C) Sigmoid activation
(D) RMSprop optimize

Answer Key

Q1	0.111
Q2	(A)
Q3	(C)
Q4	(A, D)
Q5	(D)
Q6	0.66~0.68
Q7	(A)
Q8	(C)
Q9	(D)
Q10	(C)
Q11	(D)
Q12	(C)
Q13	(A, C, D)
Q14	(A, C, D)
Q15	(A, D)
Q16	(A)
Q17	(B)

Hints & Solutions

Q1 Text Solution:

In semi-supervised learning, some examples in the dataset are labeled, while others are unlabeled, this case, the ratio of labeled to unlabeled examples is

⇒100 : 1000 - 100

⇒100 : 900

⇒1:9

⇒0.111

Q3 Text Solution:

Find the smallest non-zero distance in the matrix

Distance between x_1 and x_2= 4

Distance between x_1 and x_3 = 3

Distance between x_1 and x_4 = 6

Distance between x_2 and x_3 = 5

Distance between x_2 and x_4 = 7

Distance between x_2 and x_4 = 8

Distance between x_3 and x_4 = 8

The smaller distance is 3 between points x_1 and x_3.

Therefore, the pair of points that will be merged first in single linkage clustering.

Q4 Text Solution:

(a) Describe the distance metric, which measure dissimilarity between clusters using method like Euclidean or Manhattan distance, tailored to the specific data.

(d) Outlines the threshold for merging clusters. This threshold sets the maximum dissimilarity permitted between cluster; if it's exceed, cluster won't merge. Both are vital in hierarchical clustering, shaping cluster formation and merger based on data or cluster distances.

Q5 Text Solution:

A feedforward neural network is a type of neural network where information flows in one direction, from the input layer through the hidden layers (if present) to the output layer, without any feedback connections or cycles. This means that the output of one layer serves as the input to the next layer, and there are no loops or recurrent connections that would allow information to flow back to previous layers.

Q6 Text Solution:

We know that,

$y = f(y_{in}) = \frac{1}{1+e^{-yin}}$

The net input to the output neuron is

$y_{in} = b + \sum_{i=i}^{b} x_i\omega_i$

$= b + x_1\omega_1 + x_2\omega_2 + x_3\omega_3$

= 0.48 + (0.7 × 0.2) + (0.5 × 0.3) + (0.3 × –0.2)

= 0.48 + 0.14 + 0.15 – 0.06

y_{in} = 0.71

Put y_{in} value in Binary sigmoid activation function.

$f(y_{in}) = \frac{1}{1+e^{-yin}} + \frac{1}{1+e^{-0.71}} = 0.67$

Q7 Text Solution:

To determine the cluster centroids in the second iteration of K-Means clustering, we need to calculate the mean of the observations in each cluster. The new centroids will be:

(a): C_1: (5.3,6), C_2: (2.5,1), C_3: (8,7.5), C_4: (1.5,2.5)

Here's the explanation:

For cluster C_1 the mean of obervation (2,3) , (6,8) and (8,7) is $\left(\frac{16}{3}, \frac{18}{3}\right)$ = (5.3 , 6).

For cluster C_2 the mean of obervation (1,0) and (4,2) is $\left(\frac{5}{2}, \frac{2}{2}\right) = (2.5, 1)$.

For cluster C_3 the mean of obervation (9,9) and (7,6) is $\left(\frac{16}{2}, \frac{15}{2}\right) = (8, 7.5)$.

For cluster C_4 the mean of obervation (3,4) and (0,1) is $\left(\frac{3}{2}, \frac{5}{2}\right) = (1.5, 2.5)$.

So, the correct answer is (a) C_1: (5.3,6), C_2: (2.5,1), C_3: (8,7.5), C_4: (1.5,2.5)

Q8 Text Solution:

Explanation: Unbiasedness, in the context of statistical estimators, means that the expected value (mean) of the estimator's sampling distribution is equal to the true population parameter it is estimating. In other words, an

unbiased estimator on average provides an estimate that is equal to the population parameter.

Q9 Text Solution:

Let's analyze the statements:

(a) The chances of overfitting decrease with increasing the number of hidden nodes and increasing the number of hidden layers.

This statement Is generally false. Increasing the number of hidden nodes and hidden layers in a neural network can actually increase the risk of overfrttmg, especially If the network becomes too complex and starts fitting noise in the data it's crucial to find the right balance between model complexity and overfitting

(b) A neural network with one hidden layer can represent any Boolean function g.ven a sufficient number of hidden units and appropriate activation functions.

This statement is true A single hidden layer neural network with a sufficient number of hidden units and appropriate activation functions (such as sigmoid or ReLU) can approximate any Boolean function

(c) Two hidden layer neural networks can represent any continuous functions (within a tolerance) as long as the number of hidden units is sufficient and appropriate activation functions are used This statement Is also true. Multi-layer neural networks, including those with two hidden layers, are universal approximators and can represent continuous functions within a tolerance if they have enough hidden units and appropriate activation functions

So, the correct answer is (d) All of the above

Q10 Text Solution:

Based on the figure we can see that removing any single point would not chance the resulting maximum margin separator. Since all the points are initially classified, the leave-one-out error is zero.

Q11 Text Solution:

The formula to calculate the output size after Max pooling is given by:

$$\text{Output size} = \frac{\text{Input size - Pool size}}{\text{Stride}} + 1$$

For this case:

$$\text{Output size} = \frac{8-2}{2} + 1 = \frac{6}{2} + 1 = 3 + 1 = 4$$

Therefore, the correct answer is option (d).

Q12 Text Solution:

First calculate the output of the network:

$Z_{output} = \omega_{11} \cdot \alpha_1 + \omega_{22} \cdot \alpha_2 + \omega_{32} \cdot \alpha_3 + b_2$

Where α_i is the output of the hidden layer neuron i and $\sigma(z)$ is the sigmoid activation function.

$$\alpha_1 = \sigma(\omega_{11} \cdot x_1 + \omega_{21} \cdot x_2 + b_1)$$
$$\alpha_2 = \sigma(\omega_{12} \cdot x_1 + \omega_{22} \cdot x_2 + b_2)$$
$$\alpha_3 = \sigma(\omega_{13} \cdot x_1 + \omega_{23} \cdot x_2 + b_3)$$
$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Calculate δ_{output}:

$$\delta_{output} = (y_{output} - y_{target}) \cdot \sigma(z_{output})$$
$$\sigma'(z) = \sigma(z) \cdot (1 - \sigma(z))$$

Where,

After calculating you should find that the correct answer is option (c) δ_{output} = 0.018.

Q13 Text Solution:

Leave-One-Out cross-validation (LOO-CV) is a method where the model is trained on all samples except one, and this process is repeated for each sample in the dataset. It is an exhaustive crossvalidation technique, ensuring that each sample serves as the validation data exactly once. Option B accurately describes the LOO-CV procedure.

Q14 Text Solution:

(a) While it's true that the Softmax function outputs probabilities, it does not guarantee that the probabilities will always be in the range (0,1] due to the exponentiation involved in the calculation.

(b) The Softmax function introduces non-linearity to the network. This non-linearity is crucial for the neural network to learn complex relationships and patterns in the data.

(c) The output probabilities from the Softmax function are not independent; they are normalized probabilities, and the sum of all probabilities is equal to 1.

(d) The primary purpose of the Softmax function is to convert the raw output scores of the network into probabilities, not for regularization. Regularization is often achieved through techniques like dropout or L2 regularization.

Q15 Text Solution:

Both data points would be In a straight line If the correlation between the variables V_1 and V_1 Is 1.

As a result, all three cluster centroids will form a straight line.

Hence the Statement I is correct but Statement II is false.

K-Means Clustering is an unsupervised learning algorithm used in machine learning and data science to tackle clustering problems.

Unsupervised Learning algorithm k-Means Clustering divides the unlabeied dataset into separate clusters.

K specifies the number of pre-defined clusters that must be generated during the process; for example, if k = 2. two clusters will be created, and if k = 3, three clusters will be created.

Q16 Text Solution:

$Z = \omega_1 \cdot x_1 + \omega_2 \cdot x_2 + \omega_3 \cdot x_3$

Z= (–2·2) + (–5· 3) + (–4 ·1)

Z = – 4 – 15 – 4 = – 23

Q17 Text Solution:

The method commonly used to address the large gap between training accuracy and test accuracy, as described in the question, is Dropout.

Dropout is a regularization technique commonly used in neural networks to prevent overfitting. During training, dropout randomly deactivates a fraction of neurons in each layer. This helps the model become more robust and less reliant on specific neurons, which can be beneficial in preventing overfitting.

So, the correct answer is: (b) Dropout.